

ConnectApp Multi-Vendor Platform

Complete Technical Specifications: Integration, Workflows, & Taxonomy

1. Executive Summary

This guide defines the dashboard architecture, taxonomy data model, workflows, and key integration points between the **Admin Dashboard** and the **Vendor Dashboard** in the ConnectApp Multi-Vendor platform. The system allows seamless vendor registration, structured catalog management based on business type, commission tracking, and membership discount validations.

2. Vendor Dashboard Structure

The Vendor Dashboard adapts layout names and workflows dynamically according to the business type. Modules include:

- **Overview / Insights:** Displays real-time charts (Sales/Booking Trends, Top Selling Items, Membership Growth, Revenue by Card Tier) and summary cards (Completed/Pending Orders, Total Sales, Active Members).
- **Manage Catalog / Products:** Enables Vendors to add, update, delete, and list catalog items (products, services, doctor slots, rooms, dishes). Supports dynamic fields based on vendor type and tier pricing eligibility (Silver, Gold, Diamond).
- **Orders / Bookings:** Tracks order status (Pending, Accepted, Out for Delivery, Delivered, Checked In, Checked Out, Cancelled) and displays membership-tier applied discount levels.
- **Customers List:** Presents analytics about customers who made purchases, showing order counts and total amount spent.
- **Delivery Partners:** Enables delivery-based vendors to perform CRUD operations on their delivery riders and assign them to orders.
- **Payments & Earnings:** Shows gross earnings, platform commission deductions, and payout history.
- **Commission Overview:** Visualizes commission splits and transaction logs based on the admin-configured rate.
- **Membership Card & Scan:** Allows vendors to input or scan a Member's card QR code, validating the status, current tier, and applying matching discounts.
- **Business Settings:** Allows vendors to update business profiles, bank details, co-partner names, GST/PAN credentials, and upload licenses.

3. Admin Dashboard Structure

The Admin Dashboard acts as the central control plane of the ConnectApp platform, with access to global state and configurations:

- **Platform Insights:** Consolidated analytics across all registered members, total approval count, global sales volume, and transaction commissions.
- **Vendor Requests (Onboarding):** Lists newly registered vendors with a status of *Pending*. Admins review credentials, uploaded business licenses, and approve or reject the vendor.
- **Manage Vendors:** Directory of all approved vendors. Admins can view/update vendor details, toggle active status, or suspend/delete records.
- **Registered Members:** Global directory of all platform customers/members. Shows their active subscription tier (Silver, Gold, Diamond), card IDs, and expiry dates.
- **Payments Monitor:** A master ledger of all transaction flows, subscription renewals, and sales commissions.
- **Commission Settings:** Global controls to define and update the percentage rate the platform keeps from transactions (e.g., standard 10%).
- **Plans & Settings:** Manage subscription tiers pricing (Silver, Gold, Diamond), discount percentages (e.g., 10%, 15%, 20%), and validity periods.

4. Vendor Type, Category, and Subcategory Taxonomy

The ConnectApp platform classifies vendors using a hierarchical classification system defined in the frontend codebase at `frontend/src/data/servicesData.js`. This structure regulates dynamic registration screens, catalog forms, and frontend terminology overrides.

A. The Core Three-Tier Hierarchy

- Vendor Type (Sector):** The highest level of categorization, such as *Services*, *Products*, *Daily Needs*, *Food*, *Stay*, *Travel*, or *Jobs*.
- Category:** The second-level classification grouping related services or products together (e.g. *Electronics*, *Healthcare Services*, *Grocery*, *Restaurants*).
- Subcategory:** The specific leaf node denoting the vendor's actual business specialty (e.g., *Hospitals* under Healthcare Services, or *Smartphones* under Electronics).

B. Taxonomy Schema Mapping & UI Controls

During registration, selecting a Vendor Type dynamically narrows down the Category options, and selecting a Category filters the Subcategory choices. The frontend uses the helper function `getBaseVendorType(vendorType, category, subcategory)` to resolve the selection into a standardized base model template. The mapping works as follows:

Vendor Type (Sector)	Category	Subcategory	Mapped Base Vendor Type
Services	Healthcare Services	Hospitals, Clinics	Hospital Vendor
Services	Education Services	Schools, Colleges	Education Vendor
Products	Electronics	Smartphones, Laptops	Electronics Vendor
Products	Furniture	Sofas, Beds, Wardrobes	Home & Furniture Vendor
Daily Needs	Grocery	Rice, Wheat, Spices	Grocery Vendor
Daily Needs	Pharmacy & Healthcare	OTC Medicines	Pharmacy Vendor
Food	Restaurants / Fast Food	Fine Dining, Pizza	Restaurant Vendor
Stay	Hotels / Resorts	Budget/Luxury Hotels	Hotel Vendor
Jobs	IT Jobs	Developers, Designers	Job Vendor

5. Core Integration Workflows

A. Vendor Registration & Admin Approval Flow

1. **Registration:** Vendor fills out the register form, selecting their category/subcategory and uploading business credentials (PAN, GST, Business License). A new User document is created with role: 'Vendor' and status: 'Pending'.
2. **Admin Notification:** The request appears inside the Admin's *Vendor Requests* panel.
3. **Review & Decision:** Admin validates the vendor license and either approves (sets status: 'Approved') or rejects (sets status: 'Rejected'). Approved vendors gain backend access to API routes; others are blocked upon login.

B. Catalog Upload & Pricing Validation Flow

1. **Upload:** Vendors populate their catalog. Each item is linked to the vendor's `_id` via a `vendorId` reference.
2. **Tier Selection:** Vendors specify which membership levels can avail discount benefits for each catalog item (e.g. `cardTypes`: ['Silver', 'Gold', 'Diamond']).
3. **Admin Controls:** Admin reviews catalog statistics from the global panel to check catalog health across platform sectors.

C. Membership Verification & Discount Application Flow

1. **Presentation:** Member presents their unique QR Code (containing their Member Card ID) to the vendor.
2. **Verification:** Vendor inputs or scans the Card ID in their *Membership Card* validator. The frontend makes a request to `GET /api/vendor/verify-card/:membershipId`.
3. **Validation:** The server verifies the card is Active, checks the expiration date, and returns the membership details.
4. **Transaction:** Vendor completes the transaction, automatically applying the discount percentage linked to the Member's plan (e.g., 10% for Silver, 15% for Gold, 20% for Diamond).

D. Commission Split & Payments Settlement Flow

1. **Transaction Completion:** When an order is completed, the final amount is recorded.
2. **Commission Calculation:** The system calculates the platform commission based on the Admin's configured percentage (e.g. $\text{finalAmount} * \text{commissionRate} / 100$).
3. **Sync:** The Vendor's *Commission* tab updates with their net earnings, while the Admin's *Commission* ledger tallies platform-wide revenue cut. Payout status transitions from Unpaid to Paid when settled.

6. Critical API Endpoints Matrix

HTTP Method & Path	Authorized Roles	Description & Integration Context
POST /api/auth/register	Public	Creates new Vendor / Member accounts. Vendors default to 'Pending' status.
GET /api/admin/requests	Admin	Lists all pending vendor signup applications for approval.
PUT /api/admin/vendors/:id/status	Admin	Approves or Rejects a vendor's registration status.
GET /api/vendor/verify-card/:id	Vendor	Validates Member QR cards and returns discount percentages.
POST /api/vendor/products	Vendor	Creates a new item in the catalog linked to the vendor's profile.
PUT /api/vendor/orders/:id/status	Vendor	Updates order states, triggers commission deductions, and logs transaction audits.